

Ali Kuwajerwala

Research Engineer, AI and Robotics

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AI and robotics research engineer with a strong background in deep learning, robotics, and multimodal foundation models. Seeking technical research and engineering roles.

SELECTED PUBLICATIONS

ICRA 2024 *ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning*

Kuwajerwala, A., Gu, Q., Morin, S., et al. <https://concept-graphs.github.io>.

RSS 2023 *ConceptFusion: Open-set Multimodal 3D Mapping*

Jatavallabhula, K. M., Kuwajerwala, A., Gu, Q., et al. <https://concept-fusion.github.io>.

EXPERIENCE

Contract Research Engineer

Oct. 2025 – Present

Multiple Stealth AI & Robotics Startups

Remote

- Built policy-training infrastructure for reinforcement-learning workflows in simulation, targeting whole-body robot control and sim-to-real transfer.
- Built data pipelines for whole-body motion imitation, ingesting and normalizing large motion-capture datasets, filtering low-quality clips, and exporting training-ready datasets and data loaders.
- Developed perception and mapping software for early-stage mobile robot prototypes, including SLAM integration, sensor-processing pipelines, and field calibration workflows.

Member of Technical Staff, K-Scale Labs

Nov. 2024 – Sep. 2025

Founding Team, Deep Learning

(Palo Alto, CA)

- Led humanoid robot sim-to-real locomotion using IsaacLab², implementing rewards, actuator models, curriculum, domain randomization, logging and evaluation pipelines.
- Created K-Scale MuJoCo Viewer (KMOV)³, an interactive simulation visualizer. Built with **Qt** and **Python** for debugging policies with push forces, real-time plots, logs, and video recording.
- Created K-Infer Evals⁴, an automation system for evaluating trained locomotion policies. Runs policies in a variety of scenarios and logs results to a Notion database for comparison and inspection.
- Core contributor to the CAD → URDF pipeline, adding inertia calibration and consistency checks to keep mechanical design, simulation, and firmware in sync.

Applied Scientist Intern, Amazon

Summer 2022

Alexa AI Team, Amazon Devices

(Toronto, ON)

- Improved the accuracy of the conversational NL2SQL system by **1.5%** on the **Spider** NL2SQL dataset.
- Prototyped alternative model architectures to overcome the 512-token input-length limitation in existing models.

EDUCATION

Mila & University of Montreal

M.Sc., Computer Science (Robotics and Artificial Intelligence)

Sep. 2021 – Aug. 2024

- Supervisor:** Prof. Liam Paull, director of the Montreal Robotics and Embodied AI Lab, Core Mila Member, CIFAR AI Chair.

University of Toronto

H.B.Sc., Computer Science & Mathematics; CGPA: 3.63

Sep. 2016 – May 2020

- Award:** NSERC Undergraduate Student Research Award, **\$5,600** (2020)
- Extracurricular:** Co-Founder & Head of Operations, Robotics Club (2019–2020)
- Teaching Assistant:** Mobile Robotics (CSC477), Data Structures (CSC263), Theory of Computation (CSC236).

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, Julia, SQL, Bash

Tools / Frameworks: PyTorch, JAX, TensorFlow, MuJoCo, Isaac Sim, IsaacLab, ROS, Qt, Git, Linux